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Energy Expenditure of Organs and Tissues

Francisco Grande, MD

The basal metabolic rate (BMR) represents the sum of the metabolic activities of all the cells of the body, under the standard conditions of measurement.[1] Barcroft[2] calculated that 82% of BMR could be accounted for by the oxygen consumption of the various body organs, as determined in perfusion experiments. Field et al[3] showed that the weighed sum of the metabolic rates of rat tissues, measured in vitro, accounted for 89% of the BMR of the living rat. Martin and Fuhrman[4] reported that the summated metabolic rates of the tissues accounted for over 70% of the measured BMR of the dog in vivo, after consideration of the assumed minimal activity of the tissue slices.

The idea that BMR is proportional to the mass of "active cells" goes back to Boit and Rubner. As stated by Keys et al,[5] "Perhaps Rubner's conception of 'active tissue' is, in principle, a sound approach, although he himself was inclined to deny its importance." Rubner actually thought that body-surface area was the main determinant of BMR among the homeotherms.[6]

Several French workers have used the concept of active mass (or active protoplasmic mass). Terroine and Zunz[7] wrote that if the "active mass" determines the magnitude of BMR, then that mass should be proportional not to body weight but to body surface. But Le Breton[8] concluded that the "protoplasms" of different homeotherms are not identical. The use of BMR as predictor of the "active cell mass," as proposed by Best and Kuhl,[9] seems to be justified if we have valid information about the metabolic activity per unit of weight of this compartment. There is ample evidence to show that in the human species most of the BMR differences usually attributed to sex and age tend to disappear when the differences in fatness are considered.[10-13]

Our measurements on two groups of soldiers ($n = 13$ and $n = 12$) gave average values of 1.25 (SD ± 0.134) and 1.57 (SD ± 0.123) kcal/kg "cells"/hour. The corresponding variation coefficients were 8.8% and 7.8%.[14] Since the cells are of different tissues, and organs have widely different levels of metabolic activity, it follows that the average metabolic rate per weight unit of the cell mass will change when the proportion of tissues with different metabolic rates changes.

The Partition of BMR Among the Main Body Organs

In 1950 I tried to calculate the participation of the main organs of the human body in the BMR, using the best data available at the time. The results of this early calculation, together with some more recent estimates, are shown in Table 1. The five estimates indicate that the four organs (liver, brain, heart, and kidney) which together make up only 5% to 6% of the body weight, account for 60% to 70% of the basal metabolism. In contrast, skeletal muscle, which makes up about 40% of the body weight, accounts for only 16% to 30% of the resting metabolism.

Table 1. Various estimates of BMR partition in man, as percent of total BMR.

Organ	Ref 15	Ref 19	Ref 16*	Ref 17	Ref 18
Liver	27	26.4	33	25	32
Brain	19	18.3	16	20	21
Heart	11	9.2	11	11	10
Kidney	4	7.2	8	7	7
Subtotal 1	61	61.1	68	63	70
Muscle	20	25.6	16	30	
Subtotal 2	81	86.7	84	93	

*Recalculated from original data of Ref 16.

Oxygen Consumption by Tissue Slices

The introduction of Warburg's technique for the manometric measurement of the O_2 consumption of tissue slices gave a considerable impetus to the study of tissue metabolism in vitro. The main problem in interpreting the very extensive literature is to determine the relationship between the O_2 consumption of the tissues in vivo.

Some workers[3,4] have been able to make acceptable BMR estimates by summating the O_2 consumption values of the various body tissues as measured in vitro. Grafe et al,[20] and others, however, have expressed the view that tissue slices in vitro have a higher oxygen uptake than the corresponding tissue in vivo. These writers believe that the metabolic activity of the tissues in vivo is restricted by the influence of some regulatory mechanism operating in the whole animal. Rubner postulated that the differences in basal metabolism between large and small dogs were due not to a difference in the organization of their cells, but to the effect on their metabolic activity of impulses originated by the effect of cooling on the skin.[21]

The classic paper by Krebs[22] is an important contribution, which should be read by those interested in the problem. The effect of the medium's composition on the O_2 consumption per unit weight, or mm^3 O_2/mg tissue dry weight/hr (Q_{O_2}), of various tissues from different mammals was well illustrated by this work. Krebs's values were higher than those reported by Kleiber.[23] In fact, Krebs's values for all three media used tended to be considerably higher than the values reported in the literature using saline media[22] and of the same order as the highest values recorded using serum as the medium. Whether these values corresponded to the respiration rates of the tissues in vivo is still an open question. The oxygen uptake of dog brain-cortex slices incubated in Krebs's medium III (calcium-free) appeared to be in good agreement with values observed in vivo.[22] It

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should be noted, however, that the values used by Krebs for comparison were somewhat higher than those reported for the brain of man and monkey in vivo.[22-26]

Tissue Metabolism and Body Size

Treviranus[27] noted that the metabolic rate (CO_2 production per 100 g of body weight) was higher in small than in large animals. Sarrus and Rameaux[28] suggested that heat production should equal heat loss, which, in turn, is determined by the body's surface area. This is the origin of the so-called "surface law," which, with many supporters,[21,29-31] has dominated the physiology of BMR for more than a century. According to this view, the BMR of the homeotherms is proportional not to the weight of the body but to the body surface area, which can be calculated as a function of body weight to the two-thirds power.

After the work of Brody[32] and Kleiber,[33] it was well established that the BMR of mammals, from the mouse to the elephant or the rat to the cow, is proportional to a fractional power of body weight, usually taken as Body Weight$^{0.75}$, according to the so-called "Kleiber's rule." If we accept that BMR represents the sum of the metabolic activities of all the body cells, we then question why it does not change as a simple function of body weight to the power of one. Since cell size of homologous tissues in mammals is practically the same and remarkably independent of body size,[34] two explanations are possible: 1) that the metabolic activity of all the body cells decreases as the size of the animal increases, or 2) that the size of the "active cell mass" changes in proportion to Body Weight$^{0.75}$, rather than in proportion to the Body Weight$^{1.00}$.

Terroine and Roche[35] and Grafe et al[20] concluded that homologous tissues of different mammals respire in vitro at practically the same rate. That is, the metabolic activity of homologous tissues is independent of the size of the donor animal. Terroine and Roche stated that the metabolic rate in vitro per unit of weight of homologous tissues of large and small animals is the same, even though the metabolic rate per unit of weight of the tissue decreases in the living animal with increasing body size. Grafe et al concluded that the metabolic rate of the tissues in vivo is restricted by some regulatory mechanism to a fraction of that exhibited in vitro. It follows, therefore, that the summated metabolism of the tissues in vitro must exceed the BMR of the living animal, contrary to the results obtained by Field et al[3] and by Martin and Fuhrman.[4] It is also clear that neither Terroine and Roche,[35] nor Grafe et al,[20] interpreted their results as invalidating the surface law, but rather as proof of the fact that the tissue metabolism measured in vitro does not give a true estimate of the metabolic activity of the tissues in the living animal.

In contrast with these results are those obtained by Wels[36] and Kleiber.[33] Wels found a decreased respiration rate of the tissues with increasing size of the animal. Kleiber found a decrease in oxygen consumption by liver slices, with increasing size of the animal. He used liver slices from several rats, rabbits, and sheep, but his values for the horse and the cow were obtained on a single animal in each case. Kleiber's data showed that the Q_{O_2} decreased as a function of the animal's weight to the power 0.24. Since the metabolic rate of the whole animal is proportional to Weight$^{0.75}$, the metabolic rate per unit of weight will be proportional to Weight$^{0.75}$/Weight$^{1.00}$ = Weight$^{-0.25}$. Accordingly, these data seemed to fit with the concept that the metabolic activity of the liver in vitro changes in relation to the weight of the animal in the same way as the BMR of the animal. Krebs, however, used different media and found marked differences in the rates of respiration of a given tissue, depending on the medium used. For a given medium, the metabolic rates (Q_{O_2}) of the different tissues were affected by the size of the animal, but the influence of body size was not the same for all the tissues and was small for some.[22] The Q_{O_2} of brain cortex and kidney cortex for cattle and horse was about half of that observed for the mouse. The decrease was somewhat greater for the liver, but no one of the three tissues showed a reduction as great as that observed for the BMR per kg of body weight of the whole animal. Another way of analyzing these data is to calculate the regression of $\log_{10} Q_{O_2}$ against $\log_{10}$ body weight in grams.[37] The results of my calculations using Krebs's data[22] are shown in Table 2.

The Q_{O_2} of brain cortex and liver cortex decreased as a function of body weight to the power of 0.069 and 0.063, respectively. The liver Q_{O_2} decreased as a function of body weight to the power of 0.115, which is about half that found by Kleiber.[23]

Table 2. The relation of $\log_{10}$ tissue Q_{O_2} to $\log_{10}$ body weight (g).*

x $\log_{10}$	y $\log_{10}$	r	Regression Equation
Body weight (g)	Q_{O_2} Brain cortex	−0.969	$y = 1.621 - 0.069\,x$
Body weight (g)	Q_{O_2} Kidney cortex	−0.878	$y = 1.697 - 0.063\,x$
Body weight (g)	Q_{O_2} Liver	−0.946	$y = 1.482 - 0.115\,x$
Body weight (g)	BMR kcal/kg/day	−0.993	$y = 2.469 - 0.216\,x$

*Calculated from Krebs's data[22] for medium II. Animals measured were mouse, rat, guinea pig, rabbit, cat, dog, sheep, cattle, and horse.

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The data reported by Elliot and Henderson, analyzed in my earlier publication,[15] indicated that the Q_{O_2} of the brain cortex decreased as a function of body weight to the power of −0.1. The BMR of the whole animal per kg of body weight decreased as a function of body weight to the power of −0.216, which is close to the value of −0.25 predicted by Kleiber's rule.

Weight of the Organs and Total Body Weight

As the size of animals increases, the weight of the supporting structures tends to increase faster than body weight, and that of the internal organs increases more slowly. Thus, the data by Kayser and Heusner, analyzed by Schmidt-Neilsen,[38] indicated that the weight of the skeleton increases from the mouse to the elephant as a function of body weight to the power of 1.13. It appears then that large mammals have a greater proportion of tissues with low metabolic activity, whereas smaller mammals have a greater proportion of tissues with higher metabolic activity.

Using the data given by Brody,[32] I have calculated the regression for the combined weights of brain, liver, heart, and kidneys against the body weight for the animal, using a log-log plot. Data were used for rat, guinea pig, monkey, dog, sheep, man, hog, cow, steer, and horse. There was an excellent correlation ($r = +0.989$) between $\log_{10}$ of the summated weights and $\log_{10}$ of body weight (in grams). The regression equation was:

$$\log_{10} y = 0.825 \log_{10} x - 0.691$$

where y represents the summated weight of brain, liver, heart, and kidneys, and x the body weight, both in grams. It appears therefore that the combined weight of these organs increases as a linear function of body weight to the 0.825 power, which is close to the value 0.75.

These data seem to support the view that the decrease in BMR per unit of body weight with increasing size of the animal is due, at least in part, to the fact that the mass of organs with high metabolic rate in proportion to total body weight tends to decrease as the size of the animal increases. A similar view was postulated by Holliday et al,[18] who, however, also postulated that the decrease of BMR/kg with increasing body size may be partially due to the fact that the metabolic rate of some tissues also decreases with increasing body size.

Krebs noted that, in larger species, tissue structures accessory to the main functional cells are bound to constitute a relatively larger portion of the organs than in the homologous tissue of smaller animals. It may be that animals of different size will show little difference because, as stated by Krebs,[22] "It is to be expected that the rate of energy production of the highly differentiated cells is not dependent on the size of the animal, because the energy needed for the performance of a given piece of work is independent of the size of the body."

On the Significance of Brain Metabolism

The measurements of brain oxygen consumption in vivo reported by Schmidt et al[26] for the brain of small (3.8 kg) monkeys, and by Kety and Schmidt[25,39] for the human brain, showed that the oxygen uptake per 100 g brain/min was the same, in spite of the difference in body size of the two species. These results suggested that the metabolic rate of the brain in vivo is independent of body size. Accordingly, it would be possible to estimate the proportion of BMR that corresponds to the brain in animals of different size, using the available data for BMR and brain weight of different mammals and the same figure for the oxygen consumption of the brain. The results of this calculation are given in Table 3, making brain O_2 consumption equal to 3.5 ml/100 g/min. A similar calculation was made earlier.[15]

Table 3. Brain O_2 uptake as percentage of total O_2 consumption* (BMR) in mammalian species.

Species	Body Weight (kg)	Brain Weight (g)	Total O_2 Consumption BMR (ml/min)	Brain O_2 Uptake Total (ml/min)	Brain O_2 Uptake % of BMR
Elephant	6,650	5,700	7,880	200	2.5
Elephant	3,752	4,896	5,960	171	2.9
Horse	600	670	1,296	23	1.8
Horse	392	517	1,000	18	1.8
Pig	125	120	400	4.2	1.0
Man	70	1,400	250	49	19.6
Sheep	52	106	215	3.7	1.7
Chimpanzee	38	387	158	13.5	8.6
Dog	16.4	70	83.4	2.4	2.9
Monkey (Ateles)	3.8	91	36.0	3.2	8.8
Rabbit	2.6	19	21.7	0.7	3.1
Guinea Pig	0.800	4.7	10.3	0.16	1.6
Rat	0.250	2.0	3.4	0.07	2.1
Mouse	0.027	0.37	0.75	0.013	1.7

*Brain O_2 consumption = 3.5 ml/100 g/min for all the species.

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Man's brain takes a larger share of the total BMR (20%) than the brain of any other mammal. The monkey brain has a share of 8% to 9%, while the brains of all other mammals account for a much lower percentage of BMR. Perhaps it is not unreasonable to assume that the importance that we usually attribute to the human brain is reflected in the fact that man has to devote a large proportion of his maintenance energy requirements (BMR) to the maintenance of the brain.

The quantitative importance of brain metabolism, in relation to the total metabolism of the body at rest (BMR), is still more striking in the human infant. Holliday et al[18] calculated that, in a 10-kg infant with a 920-g brain (9% of the body weight), brain metabolism accounts for 45% of the BMR. These workers also suggested that the high BMR observed in children recovering from protein-calorie malnutrition may be partially due to the relatively large proportion of internal organs, including the brain, in relation to the total weight of the body. In one 1-year-old malnourished child with a body weight of 4.0 kg, the brain weight was 700 g, or 17.5% of the total body weight.[40]

BMR at the Molecular Level

The main implication of the "surface law" is that the thermoregulatory demands of the body are the decisive factor in determining BMR and the reason for the relationship between BMR and a fractional power of body weight, which is not far from Weight$^{0.75}$. Von Hoesslin showed that the oxygen consumption of fish was related to body surface, rather than to body weight, although the fish does not keep a constant body temperature.[41] Hemminsen[42] demonstrated that the slopes of the regression lines relating $\log_{10}$ BMR to $\log_{10}$ Body Weight for cold-blooded vertebrates, invertebrates, beech trees, and unicellular organisms are not different from those observed in mammals. As stated by Schmidt-Neilsen,[38] "This should forever dispel the notion that temperature regulation is the primary stimulus in the regular relationship between heat production and body size."

The life of the cell demands the performance of chemical processes that require a supply of energy. There is every reason to believe that BMR gives a measure of the energy cost of these processes, which are indispensable for maintaining the life and physiologic properties of all cells of the body. The partition of BMR in terms of basic, energy-demanding, biochemical processes in the resting body is the objective we must reach if we wish to understand the physiological meaning of BMR.

Two basic biochemical processes (protein synthesis and the sodium pump) seem to account for two thirds or more of the BMR. The energy requirements for whole-body protein synthesis have been estimated as accounting for 25% to 50% of BMR.[43,44] Some authors have emphasized the close correlation between whole-body protein synthesis and BMR.[44,45] But the rate of protein synthesis, like metabolic rate, changes from organ to organ. Therefore, we should like to know the participation of the individual organs in the protein synthesis of the whole body. Some organs with high metabolic activity, like the brain, are not the most active centers of body protein synthesis. It is likely that most of the brain's energy consumption is used for active transport of ions to sustain and restore the membrane potentials discharged during the process of excitation and conduction.[24] Other organs, like the kidney, seem to require energy to perform processes such as sodium reabsorption.[46]

The task ahead is, therefore, to identify the basic energy-demanding processes that account for the BMR and to determine the quantitative participation of the various organs and tissues in such processes.

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Current Status of Nonshivering Thermogenesis

Jean Himms-Hagen, PhD

Definition of Nonshivering Thermogenesis

Nonshivering thermogenesis is an increase in metabolic rate, over and above the basal metabolic rate (BMR), which occurs in response to reduced ambient temperature and which is not due to mechanical activity of muscle. It is a regulatory[1,2] or facultative component of the total metabolic rate, as opposed to the obligatory component or BMR; it diminishes to zero when an animal is at its thermoneutral temperature and is analogous to shivering thermogenesis, which is likewise a facultative process. Nonshivering thermogenesis can be distinguished from thyroid-induced thermogenesis in that it can be quickly switched on and off according to the rapidly changing needs of the animal. A change in thyroid-induced thermogenesis, on the other hand, involves a change in obligatory thermogenesis.

Occurrence of Nonshivering Thermogenesis

Greater or lesser degrees of nonshivering thermogenesis can be observed in the same animal at different times, depending on the ambient temperature and on the species and age of the animal. Hibernators, such as the hamster, usually have a large capacity for nonshivering thermogenesis, and this process is their principal source of heat, not only during exposure to cold but also during arousal from hibernation. Adult nonhibernators usually have a small (eg, rat, guinea pig, mouse) or virtually absent (eg, sheep, man) capacity for nonshivering thermogenesis, although the newborn human generally can use this process when required. Some nonhibernators (eg, rat, guinea pig, mouse) can adapt to living at low temperatures by increasing their capacity for nonshivering thermogenesis. Such an increase in capacity is a slow adaptive process, which requires 2 to 3 weeks for its full development.

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